

Absolute continuity of non-Gaussian and Gaussian multiplicative chaos measures

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Discrete Random Structures II
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Based on joint work with Yujin H. Kim (Caltech)

Outline

- ① Background on GMC
- ② Main result
- ③ Construction of the coupling (outline)
- ④ Future directions

Background on GMC

Goes back to Kahane (1985). Study of

$$“M_\gamma(dt) = e^{\gamma X(x) - \frac{\gamma^2}{2} \text{Var}[X(x)]} dx,” \quad (1)$$

where $(X(x))_{x \in D \subseteq \mathbb{R}^d}$ is a centered, log-correlated Gaussian field, $\gamma \in \mathbb{R}$, i.e.

$$\text{Cov}[X(x), X(y)] = \log(1/|x - y|) + g(x, y),$$

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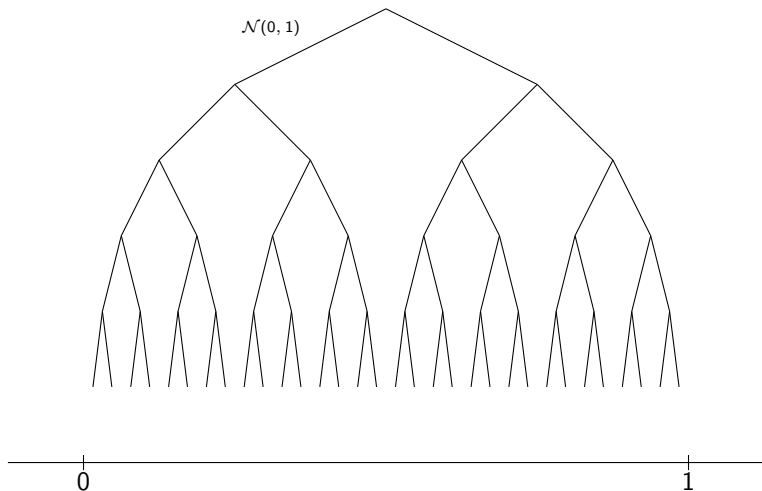
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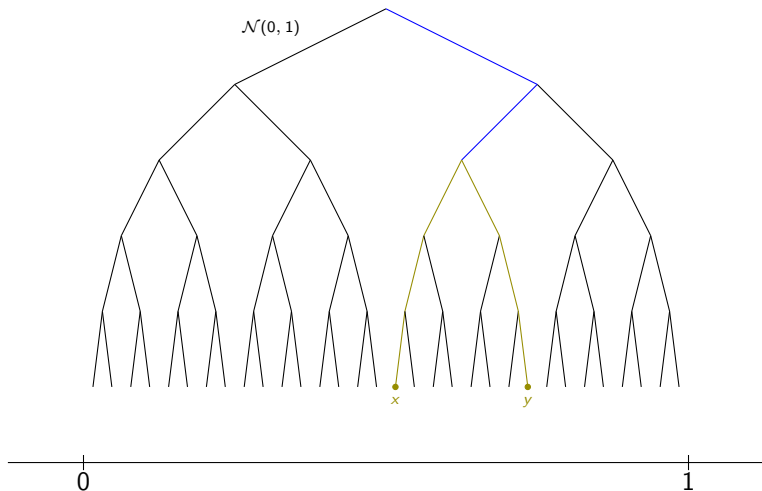
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Branching random walk



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Consider $S_n(x) = \sum_{k=1}^n \frac{g_k \cos(2\pi kx) + g'_k \sin(2\pi kx)}{\sqrt{k}}.$

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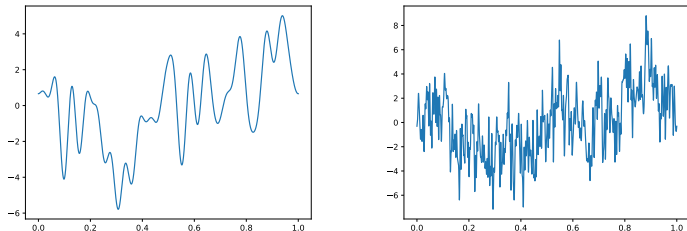


Figure: $S_n(x)$ for $n = 20$, $n = 200$

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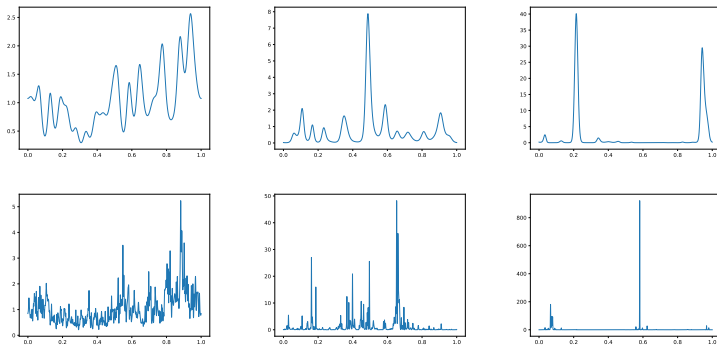


Figure: $e^{\gamma S_n(x) - \gamma^2/2 \text{Var}[S_n(x)]}$ for $n \in \{20, 200\}$, $\gamma \in \{0.2, 0.8, 1.4\}$.

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- M_γ is non-trivial for $\gamma \in (0, \sqrt{2d})$
- Approximation doesn't matter (Kahane '85, Robert-Vargas '10, Shamov '16, N. Berestycki '17)

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- for $\gamma \in (0, \sqrt{2d})$ there exists $0 \neq \mu_{\gamma,a}$, s.th. $\lim_{n \rightarrow \infty} \mu_{n,\gamma,a} = \mu_{\gamma,a}$ almost surely (in topology of weak convergence)
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Can we couple $(a_k^{(i)})_k$ to $(g_k^{(i)})_k$ i.i.d. standard Gaussians s.th. the chaos measures are mutually absolutely continuous?

Main result

Theorem (Kim-K., '24)

Take $\gamma \in (1, \sqrt{2})$. There exists an i.i.d. sequence of standard Gaussians $(g_k^{(i)})_{k \in \mathbb{N}, i \in \{1,2\}}$ coupled to $(a_k^{(i)})_{k \in \mathbb{N}, i \in \{1,2\}}$ such that $\mu_{\gamma,g}$ and $\mu_{\gamma,a}$ are almost surely mutually absolutely continuous.

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- $\mu_{\gamma,g}$ is a GMC.
- Result generalizes to fields of the form

$$\sum_{k=1}^n a_k \lambda_k^{-\frac{d}{4}} e_k(x),$$

where a_k are i.i.d., $\{e_n\}_{n \in \mathbb{N}}$ is an orthonormal basis of eigenfunctions of $-\Delta$ on D with Dirichlet boundary conditions in increasing order of eigenvalues λ_n .

- In this setting $\gamma \in (\sqrt{d}, \sqrt{2d})$.

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Same strategy, additional input is a new high-dimensional CLT proven by Chowdhury-Ganguly '25.

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- Turns out: Still too many points. Crucial step: $\mu_{\gamma,a}$ is determined by $2^{n-\frac{\gamma^2}{2}n}$ points. This many points can be coupled via Yurinskii '78 for $\gamma > 1$ or via Chowdhury-Ganguly '25 for all $\gamma > 0$.

Future directions

- More general fields.
- Use the coupling to construct the critical chaos (WIP).
- Use the coupling to show that some other approximations still give the same limit (WIP).
- Use the coupling to transfer any other properties from GMC to the non-Gaussian case.

Thanks:)

Thanks for listening

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- Set $\tilde{S}_n(x) := \sum_{k=1}^m \tilde{a}_k(x)$, piecewise constant on $[t_m(v), t_m(v+1))$, we also write $\tilde{S}_n(v)$, $v \in \{1, \dots, 2^{(1+\delta)n}\}$.

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- Roughly: Replaced field by $\approx$ binary cascade. Caveat: no innergenerational independence

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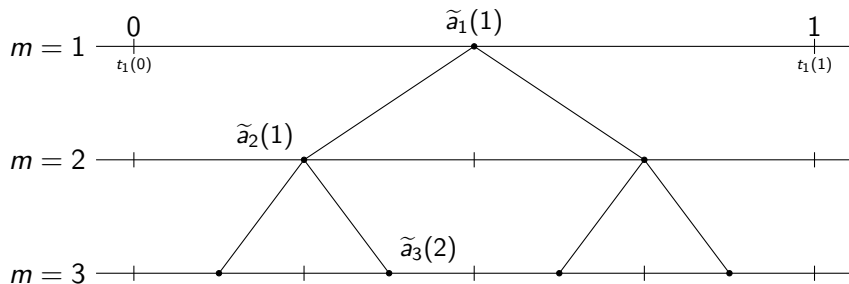


Figure: Discretization up to level 3, we have $\tilde{S}_3(2) = \tilde{a}_1(1) + \tilde{a}_2(1) + \tilde{a}_3(2)$.

We set $\tilde{\mu}_{\gamma,a} := \lim_n e^{\gamma \tilde{S}_n(x)} / \mathbb{E}[e^{\gamma \tilde{S}_n(x)}] dx$ and show

$$\sup_{n \in \mathbb{N}} \sup_{x \in [0,1]} |S_{2^n-1}(x) - \tilde{S}_n(x)| < \infty.$$

This means that $\tilde{\mu}_{\gamma,a}, \mu_{\gamma,a}$ are almost surely mutually absolutely continuous.
Method: Chaining. Independent of γ .

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Recall: We have increments

$$\tilde{a}_m = \left(\sum_{k=2^{m-1}}^{2^m-1} \frac{a_k^{(1)}}{\sqrt{k}} \cos(2\pi k t_m(v)) + \dots \right)_{v \leq 2^m}.$$

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But: $\tilde{\mu}_{\gamma,a}[\mathcal{T}_{m,\gamma}^c] \rightarrow 0$, where

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Yurinskii coupling (for us) gives that for $D \subseteq \{1, \dots, 2^m\}$ there is $(\tilde{g}(v))_{v \in D}$, such that $\text{Cov}[\tilde{g}] = \text{Cov}[\tilde{a}]$ and

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Recall: We have increments

$$\tilde{a}_m = \left(\sum_{k=2^{m-1}}^{2^m-1} \frac{a_k^{(1)}}{\sqrt{k}} \cos(2\pi k t_m(v)) + \dots \right)_{v \leq 2^m}.$$

We can't couple the full vector to a Gaussian vector, 2^m points vs. 2^m summands.

But: $\tilde{\mu}_{\gamma,a}[\mathcal{T}_{m,\gamma}^c] \rightarrow 0$, where

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Thus (by taking δ small enough) we are good once $1 - \frac{\gamma^2}{2} < 1/2$, i.e. $\gamma > 1$.

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Chowdhury-Ganguly '25 develop a coupling adapted to the covariance to replace the Yurinskii coupling, gives all $\gamma \in (0, \sqrt{2})$.

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We coupled $\tilde{S}_{m,a}$, $\tilde{S}_{m,g}$, $m \in \mathbb{N}$ such that

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Then:

$$\mu_{\gamma,a}[A] = \lim_{m \rightarrow \infty} \int_A R_{m,\gamma}(x) \tilde{\mu}_{\gamma,g}(dx) = \int_A R_{\infty,\gamma}(x) \tilde{\mu}_{\gamma,g}(dx).$$

And we are done.